Prosodic Typology: Stress, Accent, Tone, and Intonation
Predictable Stress and Foot Structure

Vertiefung Indoiranisch: Indoiranische und indogermanische Prosodie

21.11.2023

Schedule (Tagesordnung)

1. **Readings:**
   - This Week: **Féry 2017: Ch. 7**, **Kager 1999: Ch. 4**, Hayes 1995: Ch. 3, Hyman 2009
   - Next Week: **Macdonell 1910: 76–82**, **Beguš 2016**, Gunkel 2017, Witzel 2020
2. Revisiting Prosodic Typology: Stress, Tone, and Intonation
3. Analyzing Stress Assignment I: Basic Typology
4. Analyzing Stress Assignment II: Interactions in OT

1 Revisiting Prosodic Typology: Stress, Tone, and Intonation

1. Proposed three-way typology of languages based on their use of fundamental frequency (F0, pitch):

   1. 1. Intonation-only: Pitch is derives from melodic contours that are employed, on the one hand, to grammatically express information about discourse and information structure (e.g., question, focus), and on the other, to mark prosodic constituents above the level of the prosodic word; intonational "tunes" may be mapped onto or spread across phrases or entire utterances (cf. Gussenhoven 2004: 22).
      - Such languages may be canonical stress languages (e.g., English, Russian) or simply lack word-level stress or pitch altogether (French, perhaps Korean).

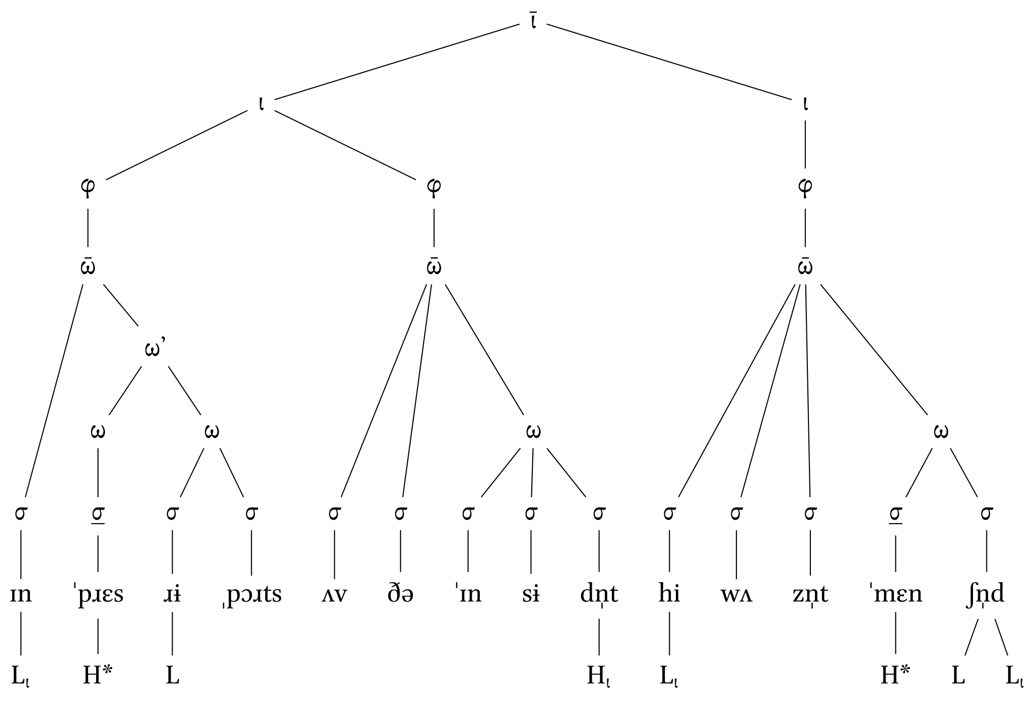
   1. 2. Tone: Lexical entries bear information on specific F0 excursions in combination with segments, and the F0 excursions realized on the surface are not attributable primarily to pure intonation.
      - Canonical tone languages usually have relative "dense" lexical tone, where contrastive lexical specifications are possible, minimal pairs are abundant, and the realization of tone is largely orthogonal to other aspects of prosody, include Iau (Lakes Plain, West Papua; Bateman 1990), Thai, Mandarin Chinese, and Vietnamese.

   1. 3. "Pitch-Accent": F0 excursions consistently occur at the level of the prosodic word (ω) in a typified pattern. Lexical pitch distinctions are sparse or non-existent. The rules determining where pitch excursions occur in the prosodic word can look very similar or identical to rules used for assigning stress.
      - E.g., Cairene Arabic assigns stress (prosodic prominence) via construction of feet in a fully predictable way, and a High (H*) pitch associates with *every* stressed syllable (Hellmuth 2006, Hellmuth 2007).

2. More on **intonation**

- Standard model used is the so-called "Autosegmental Metrical Model": see Pierrehumbert 1980 (in large part inspired by the earlier work in Liberman 1975 and Bruce 1977), Ladd 1996 for a detailed introduction to the topic, or Arvaniti and Fletcher 2020 or Féry 2017: Ch. 5 for a briefer overviews.
- AM treats intonation through the interaction of an autosegmental tier for melodies with metrical structure and prosodic constituents. The AM model assumes that languages possess a lexicon of tonal morphemes ("tunes") that contribute to the semantic and/or pragmatic interpretation of an utterance.
- Phonologically, the tunes consist of (sequences of) tones specified for pitch height and alignment with segmental material. These tunes generally consist of Low (L) and High (H) autosegments (tonemes), though some analysts also include Mid (M) tones, which may further be specified to associate at or near "accented" tone-bearing units or at the edges of prosodic constituents.
  - The former are standardly indicated with * (e.g., H*); these are **intonational pitch accents**.
  - The latter are standardly marked with are preceding or trailing % (e.g., L%); these are boundary tones, which may be more precisely indicated as aligning with the boundary of particular types of prosodic constituents (e.g., L_φ).
- Note that intonational pitch accents (starred tones) must associated with stressed syllables.
- Although *duration* is the primary phonetic correlate of stress in most varieties of English (Gordon and Roettger 2017, van Heuven 2019, van Heuven and Turk 2020), the fact that pitch accents always fall on some (though not all) stressed syllables makes English speakers also sensitive to F0 in the perception of stress.
- Example (1), borrowed from Gussenhoven 2004: 23, is an utterance comprised of two intonational phrases, each of which carries a declarative intonation (cf. Gussenhoven 2004: 296–8).

(1) Declarative Intonation: *In press reports of the incident, he wasn't mentioned.*



3. More on **tone**

- A language should be said to possess tone if some of its phonological lexical entries must bear contrastive information (at least privatively) about pitch.
- Canonically, such a contrast consists in a distinction of at least two types of pitch.
- Example (2) reproduces a typical set of minimal pairs for pitch in Ibibio, which underlyingly distinguishes a High (H) and a Low (L) tone, and which may combine into a surface falling contour tone; any TBU (a vocalic mora) may bear either tone, unpredictably. See similarly McCawley 1978 or Yip 2002 for this phonological view of tone as based on lexical specification of pitch in the phonological entry of morphemes.

(2) Contrastive Lexical Tone in Ibibio (Niger-Congo, Nigeria)[1]
   a. H–H: /ák͡pá/ → [ʔák͡pá] 'expanse of ocean'
   b. L–L: /àk͡pà/ → [ʔàk͡pà] 'small ant species'
   c. H–L: /ákù/ → [ʔákù] 'priest'
   d. L–H: /àkú/ → [ʔàkú] 'first'
   e. H–HL: /ák͡pân/ → [ʔák͡pân] 'square woven basket'
   f. L–HL: /àk͡pɔ̂/ → [ʔàk͡pɔ̂] 'rubber tree'

- Sparsity of surface pitch distinctions, where phonetically only High and Low are observable, and phonologically at most a privative distinction between a lexically specified tone and a default tone would be required, are hardly uncommon.
- Most Athabaskan languages, for instance, exhibit only a surface H vs. L distinction (Krauss 2005), as do many varieties of North Germanic languages (Oslo Norwegian, Stockholm Swedish), Seneca (Iroquoian; Chafe 1977, Melinger 2002), Chimwiini (Bantu; cf. Kisseberth and Abasheikh 2011) varieties of Basque (Hualde 2000, Hualde 2012), and most varieties of Japanese (including the most well-described dialect, that of Tōkyō).

⇒ Are these languages with tone, or are they languages with "pitch accent", or do they have both?

4. More on **pitch accent**

4.1. Larry Hyman has argued in a series of publications that "pitch accent" is not a meaningful category, being used to describe many languages with many distinct types of prosodic systems (Hyman 2006, Hyman 2009, Hyman 2014).

- From Hyman's perspective, languages that conform to the prototype of "stress" all exhibit the following phonological properties of word-level prosody: CULMINATIVITY, OBLIGATORINESS, SYLLABLE-ORIENTATION, PRIVATIVITY, and METRICALITY.

(3) Minimum Properties of Languages with Stress (after Hyman 2014: 60–1)
   a. SYLLABLE-DESIGNATION: the stress-bearing unit is the syllable (cf. Hayes 1995: 49–50)
   b. OBLIGATORINESS: every lexical word has at least one syllable marked for the highest degree of metrical prominence (= primary stress).
   c. CULMINATIVITY: every lexical word has at most one syllable marked for the highest degree of metrical prominence.

[1] Data and recordings available from the UCLA Phonetics Lab at `http://www.phonetics.ucla.edu/appendix/languages/ibibio/ibibio.html`.

d. PRIVATIVITY: a given a given foot/syllable/mora either has prominence or lacks it, such that the prosodic constituent may be labeled either "stressed" or "unstressed".
Pitch may similarly be treated privatively, if TBUs need only be specified as bearing one type of pitch, e.g., only an underlying contrast between /H/ or /Ø/.
e. METRICALITY: prosodic prominence is positionally restricted (typically by reference to foot structure).

- Some languages may lack one or more of these properties. Such languages are frequently given the label "pitch-accent language", though they may differ in crucial respects.
- Tōkyō Japanese lacks OBLIGATORINESS.
- Seneca (Iroquoian) lacks both OBLIGATORINESS and CULMINATIVITY, though the assignment of high tone seems dependent on foot structure, like Cairene Arabic
- Note that Vedic Sanskrit and Ancient Greek seem very close to Hyman's phonological stress prototype: almost every word has one and no more than one high pitch (OBLIGATORINESS, CULIMINATIVITY), and at least in Vedic, the element bearing the high pitch appears to be the syllable.

4. 2. To briefly illustrate the type of issues involved, consider the analysis of the distribution of H in Seneca offered by Melinger 2002 (cf. also Hyman 2009: 227–8).

- High tone occurs on the first syllable of every syllabic trochee (built starting from the peninitial syllable, left-to-right), provided that at least one of the syllables in the foot in closed.
- Insofar as pitch peaks in prosodic words Seneca are fully predictable based on metrical structure, the lexical specification requirement for tone is not satisfied.
- The fact, however, that more than one H is allowed, or no H is required, contrasts with the canonical enforcement of CUMULATIVITY and OBLIGATORINESS in stress systems.

(4) Predictability of the Distribution of H in Seneca
a. One H (leftmost foot has a closed syllable): [oʔ.(dís.wa).(de.njẽ:).doh] 'you (PL) waded'
b. One H (rightmost foot has a closed syllable): [a.(gẽ.ni).(yás.da).yeʔ] 'I'm willing'
c. Multiple H (first and third feet have closed syllables): [de.(wá.geʔ).(nyo.da).(gẽ́.ʔõh)] 'I'm busy'
d. No H (all feet contain only open syllables): [de.(ga.de).(nye.o).dẽʔ] 'I'll put a necktie on'

4. 3. The prosody of Lithuanian (Kenstowicz 1972, Young 1991, Blevins 1993, Kushnir 2018) illustrates a radically different case.

- The position of stress is unpredictable, and the realization of tone is dependent upon stress.
- In heavy syllables of the shape containing [V:] or [VR], if under the stress, one of two distinct pitch contours may be realized.
- See 5, where preceding ˈ marks stress and a circumflex or acute represent distinct tones (the traditional "circumflex" represents a sustained or rising tone, while the traditional "acute" represents a falling tone). [The phonetics of tone in Lithuanian are in fact quite unclear (see Dogil 1999), but most likely, the "circumflex is *not* a rising tone.]

(5) Lituanian Stress–Tone Interaction
a. *vaˈdõvas* [vaˈdǒ:vas] 'leader:M.NOM.SG'
b. *vadoˈvu* [vado:ˈvu] 'leader:M.INST.SG'

c. *ˈkálti* [ˈkaÎti] ‘to forge’
d. *ˈkal̃tas* [ˈkǎltas] ‘guilty:NOM.SG.M’

- Note that, when stress moves away from a syllable specified for tone, the tone does not surface; as the near-minimal pair between 5c. and d. indicates, the distinction between rising and falling tone is best treated as lexically specified.
- But insofar as all stressed heavy syllables exhibit one or the other tone, the surface distinction may be treated as phonologically privative, i.e. /Marked Tone/ vs. /∅/.
- The fact that these pitch contours are unpredictable necessitates their (partial) lexical specification, and thus principal criterion for tone (lexically specified pitch features) is fulfilled.
- The “unmarked tone” could then be a default word-level pitch-accent, perhaps H*L. Under this analysis, words with falling tone (“acute”) would have no lexical tone marking, but stress on a long syllable, while words with sustained H tone could have a lexical /H*/ associated with some syllable. Both the lexical /H*/ and the default pitch accent H*L would appear identical when stress occurs on a light syllable.

4. 4. The situation in Stockholm/Central Swedish (and other North Germanic varieties) is very similar.

- Traditionally, C. Swedish is described as having a contrast between “accent 1” and “accent 2” (”acute” and ”grave” in older literature).

(6) Swedish Accent 1 vs. Accent 2[2]

a. [¹ˈand-en] ‘duck-the’ [²ˈande-n] ‘spirit-the’
b. [¹ˈsteg-en] ‘steps-the’ [²ˈstege-n] ‘ladder-the’

- On the account of Riad (2014), the distinction is due to lexical marking of tone for accent 2, while accent 1 simply reflects default intonation (pitch accent at the level of the ω).
- Accent 2 must be realized on the syllable with primary stress, and must be followed by at least one further syllable; the consequence of these two restrictions is that accent 2 may reassociate from a host morpheme to a syllable in another morpheme that receives the primary stress, and that accent 2 is never found on monosyllables or on forms with ultimate stress.
- The basic tonal patterns or melodies are H*L for accent 2, but HL* for accent 1. Under focus, the fundamental difference – just a lexically specified H* – becomes clear, since Accent 1 shows L*H, but Accent 2 H*LH; under focus, both accentual types contain the sequence LH (which is then also followed by L% at the end of a word), but in accent 2 forms, a H* precedes the LHL% sequence.

4. 5. Given this overall picture, the following definition of pitch accent languages seems attractive: these are languages with a default intonational contour at the level of the prosodic word, where the “pitch accent” (starred tone) associates to the syllable with primary stress.

- Stress – understood as metrical structure – might otherwise have no other phonetic correlates! See Bennett 2012 on segmental evidence for foot structure in languages that otherwise provide no phonetic indications of stress.
- In languages like Egyptian Arabic (and probably Vedic Sanskrit), stress is obligatory, and a default pitch accent is nearly always realized on a stressed syllable.
- In languages like Seneca (and probably Tōkyō Japanese), stress is not obligatory, and the default pitch accent is only realized if the word has stress.
- In languages like Lithuanian or Central Swedish (and probably Ancient Greek), stress is obligatory, and there is both a default pitch accent *and* limited contrast in lexical tone.

[2]Historically, the tonal word accent of “Accent 2” found in some North Germanic (Central Swedish, Eastern Norwegian) varieties is related so-called *stød* (a prosodically distributed creaky voice or glottal stop) in other varieties (Standard Danish).

2 Analyzing Stress Assignment I: Basic Typology

5. In many languages, stress is non-contrastive and completely predictable based on surface phonological structure: there are no minimal pairs of lexical items distinguished solely by accent.

(7) **Phonologically Predictable Strss**
   a. Hungarian: stress on leftmost syllable, always.
   b. Latin: stress on a heavy penult, else on the antepenult.
   c. Tocharian B: stress on the second syllable from the left, unless stress would be on the final syllable (with some lexical exceptions, and opacity in derivations)
   d. Polish: stress accent on the penult (with some lexically exceptional Latinate borrowings, e.g., *uniwérsytet*)
   e. Kich'e (Mayan) : stress accent on the ultima.

- Languages of type 1 can usually have their accentuation, including any secondary stress, determined by reference to foot structure and/or boundaries of the prosodic word.
- Assumption: stress assignment makes reference to other prosodic structures (syllables and feet).
- In short, such languages need only refer to sequences of segments, and the algorithms of syllabification, footing, and stress assignment take care of the rest.

6. General principles:

- Construct feet based on syllables and syllable weight.
- Feet have "strong" and "weak" constituents when they consist of more than one syllable (i.e., are "iambic" or "trochaic")
- Assign word-level (primary stress) prominence to some particular foot.
- Assign secondary stress to any other feet.

7. Typology of Feet in Hayes 1995:

- Syllabic trochees: feet are not sensitive to syllable weight. A foot consists of two syllables, and the leftmost syllable in the foot is the "strong" member.
- Moraic trochees: feet are sensitive two syllable weight. A foot consists of two moras, and the leftmost syllable in the foot is the "strong" member (when a foot contains more than one syllable).
- Iambs: feet are partially sensitive to syllable weight. A foot consists of either two syllables in which the first has one mora, or a single bimoraic syllable. The rightmost syllable in the foot is the "strong" member.

8. **Examples of foot types:**

8.1. Pintupi (Pama–Nyungan): *syllabic trochees*. Mark the leftmost trochee as the most prominent. Note alternating pattern of primary and secondary stress.

(8) Pintupi Accent Assignment
   a. (pá.ɳa) 'earth'
   b. (tʲú.ʈa).ja 'many'
   c. (má.ɭa)(wà.na) 'through from behind'
   d. (pú.ɭiŋ).(kà.la)tʲu 'we sat on the hill'

e. (tʲá.mu).(lìm.pa).(tʲùŋ.ku) ‘our relation’

8. 2. Latin: *moraic trochees* with final syllable extrametricality. Mark the rightmost moraic trochee as the most prominent.

(9) Classical Latin Accent Assignment

a. Antepenult: nom.sg. (sá.pi.)⟨ēns⟩

b. Penult: nom.pl (sa.pi.)(én.)⟨tēs⟩

c. Antepenult, with unparsed penult (no degenerate feet permitted): cmpv.nom.sg. (sa.pi.)(én.)ti.⟨or⟩

d. **False**: cmpv.nom.sg. ^{X}(sa.pi.)(en.)(tí.or)

e. Penult: acc.sg. (ví).⟨rum⟩

f. Penult: vi(rúm)⟨que⟩

8. 3. Often, the construction of foot structures may be implicated in a wide array of phonological and morphophonological processes. See Mester 1994 for detailed evidence as to how moraic trochees can affect numerous phenomena in Latin.

8. 4. Kich’e:

(10) Kich’e Accent Assignment: iambs of the shape (⌣ x) or (—); mark rightmost iamb as most prominent; vowel lengthening and apocope avoid unfootable/degenerate light syllables

a. (ka.mík) ‘today’

b. (kìn)(b’íj) ‘I tell’

c. /tzijob’al/ → (tzì:).(jo.b’ál) ‘story’

d. /ranima/ → (rà:).(nimá) ‘his/her heart’

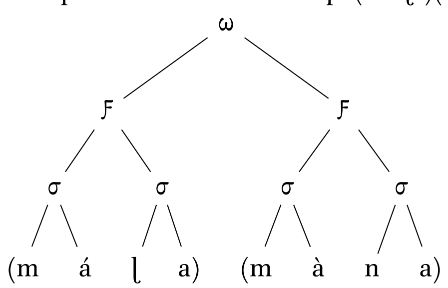
e. (u.tì).(ni.mít) ‘his/her village’

f. /kapetik/ → (kpe.tík) ‘(s)he travels’

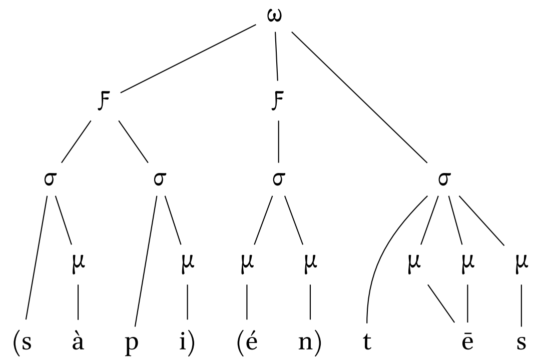
g. (kìn).(cha.kún) ‘I work’

h. (kàx).(làn).(tík) ‘Spanish’

8. 5. Example Prosodic Tree: Pintupi (má.ɭa)(wà.na) ‘through from behind’



8. 6. Example Prosodic Tree: Latin nom.pl (sà.pi.)(én.)⟨tēs⟩



9. **Segment Extrametricality and Extrasyllabicity**

9. 1. Segment extrametricality and extrasyllabicity refer to the seeming “invisibility” of certain units to the parsing of those units into higher-level prosodic structures.

9. 2. Extrametricality and extrasyllabicity seem only at the **edges** of prosodic words.

9. 3. We can identify two we can identify two common types of extrametricality:

   1. Segment extrametricality: segments are not parsed into moras ∼ are not assigned a mora through Weight-by-Position.
   2. Syllable extrametricality: syllables are not parsed into feet.

9. 4. Segment extrametricality and syllable extrametricality is common at the right edge of the word. The extrasyllabicity of segments may occur at both the left and right edges of a prosodic word (see Indo-European evidence in Byrd 2015).

9. 5. As seen above, Latin has syllable extrametricality: the rightmost syllable in a word is ignored in foot formation. In this way, we can obtain a nice generalization about Latin stress: the rightmost foot in the word is assigned the primary stress.

9. 6. Extrasyllabicity and Extrametricality are different types of the same phenomenon!

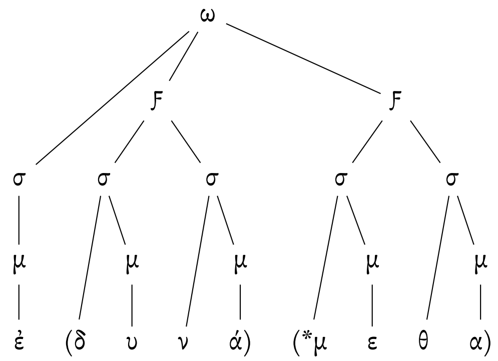
   - Extrasyllabic segments are segments that are not parsed into syllables.
   - Extrametrical units are either segments that are not assigned a mora, or syllables that are not parsed into feet.

9. 7. Final segment extrametricality is necessary to coherently express the assignment of the “recessive” accent in Attic-Ionic Greek (the * represents the stressed syllable, which receives the L* portion of a HL* tone, with the result that the H portion, marked with acute, stands one syllable farther to the left).

(11) Attic-Ionic Default Ictus Assignment

   a. nom.sg. (πο.λυ)(άν.)(*θρακ)⟨ς⟩ ‘having much coal’
   b. nom.sg. (ἄν.)(*θρω.)πο⟨ς⟩ ‘man’
   c. acc.pl. (ἀν.)(θρώ.)(*που)⟨ς⟩
   d. 1.sg.impf.mid.ind. ἐ.(δυ.νά.)(*μη)⟨ν⟩ ‘I was able’
   e. 1.pl.impf.mid.ind. ἐ.(δυ.νά.)(*με.θα) ‘we were able’
   f. **False**: nom.sg. ^{X}(ἀν.)(θρώ.)(*πος)

9. 8. Example Prosodic Tree: Attic 1.pl.impf.mid.ind. ἐ.(δυ.νά.)(*με.θα)



10. **Word Minimality and the Foot**

10. 1. The central idea: since a prosodic word should dominate at least one foot, if a foot is required to be minimally disyllabic or bimoraic, then a language might ban words of the shape /CV̌/ or /CV̌⟨C⟩/, perhaps repairing them by vowel lengthening.

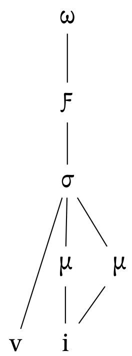
10. 2. The evident ban on monosyllabic /(C)V̌/ words in Latin (discussed *inter alia* by Kuryłowicz 1968: 191, Allen 1972: 51, Golston 1991: 101–23, Mester 1994: 20–23, Prince and Smolensky 1993 [2004]: 49–50, and Byrd 2015: 113–7), clearly holding for at least lexical words, exemplies just such a typical minimality requirement.

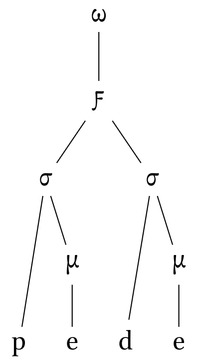
10. 3. Golston (1991) offers the most detailed discussion of word minimality in Latin, and argues in favor of extrametricality of word-final obstruents. Under this account, Latin has function words of the shape /CV̌C/ that are not obliged to satisfy a minimal word requirement, while all instances of lexical words that surface as [(C)V̌P] (e.g., *lac* 'milk:NOM/ACC.SG', *os* 'bone:NOM/ACC.SG') are argued to be underlyingly /CVCC/ and undergo a post-lexical cluster simplification, after the prosodic word has already been constructed.

(12) Minimal Prosodic Words in Latin

a. Well-Formed Bimoraic Feet (/(C)V:/, /(C)V̌CV̌/)

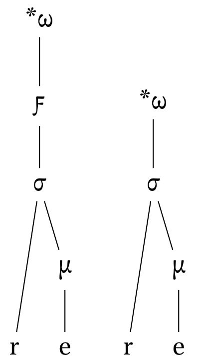
'force:ABL.SG' 'foot:ABL.SG





b. No Degenerate Monomoraic Foot or Violation of Headedness

*F = μ   Headedness



10. 4. The presentation of monosyllabic lengthening in Byrd 2015: 113–7 is confusing, because it treats monosyllabic lengthening as evidence for extrasyllabicity of word-final segments.

10. 5. It is actually better evidence for *word-final consonant extrametricality*.

10. 6. Ponapean monosyllabic lengthening is due to final-segment extrametricality:

(13) Ponapean Monosyllabic Noun Lengthening.

a. /pik/ → [piː⟨k⟩] 'sand' (vs. inflected *pik-en* 'sand of')

b. /pet/ → [peː⟨t⟩] 'bed' (< Eng. *bed*)

c. /keːp/ → [keː⟨p⟩] 'yam', not **[keːː⟨p⟩]

d. /kent/ → [ken⟨t⟩] 'urine', not **[keːn⟨t⟩]

10. 7. Greek monosyllabic nouns show lengthening due to final-segment extrametricality, too!

(14) Attic-Ionic Monosyllabic Noun Lengthening

a. Nom. [póːs] vs. gen. [podós] 'foot' (/pod/)

b. Nom. [mû̂s] vs. dat.pl. [musí] in 'mouse' (/mus/)

3 Analyzing Stress Assignment II: Interactions in OT

11. We can illustrate how some constraints interact to derive predictable metrical stress systems using Tocharian B as a case study.

- Tocharian B (also known as West Tocharian) is, alongside Tocharian A (or East Tocharian) one of the representatives of the Tocharian branch of the Indo-European family. The Tocharian languages are attested from the period approximately 500–1000 CE in the Tarim Basim (Xinjiang, China) primarily in the form of Buddhist literature translated from Sanskrit.
- The fundamental generalizations concerning non-exceptional stress assignment in Tocharian B (TB) have been known since Krause 1952: 10 (cf. also Krause and Thomas 1960: 42–7, Pinault 2008, Malzahn 2010: 5). Burroni (2017) has provided an analysis of just aspects of phonologically predictable stress following the framework of Gordon 2002; Koller (2021) has recently subjected stress assignment in Tocharian B, including its morphologically conditioned aspects and interaction with other phonological processes, to a rigorous analysis in the parametric framework of Hayes 1995.

- The presentation here draws essentially on the data and analysis provided in the unpublished work of Burroni and Koller. See also Kim 2011 for discussion on the prehistory and diachrony of stress in Tocharian.

12. The position of stress in a given word can often be established on the basis of stress-conditioned allophony of the central vowel phonemes /ə/ and /a/, represented in (15).

(15) Allophony of Central Vowel Phonemes in Tocharian B

	/ə/	/a/
[ɨ]	[ə]	[a]
= ⟨ä⟩	= ⟨a⟩	= ⟨ā⟩

- Since [ɨ] ⟨ä⟩ never occurs under the primary stress, while [a] ⟨ā⟩ is found exclusively when bearing primary stress, syllables with one of these vowels as its nucleus must be unstressed or stressed, respectively.
- Depending upon the pattern of alternation in which a given vowel in a given root or affix is implicated, its underlying quality can be established, as well as the position of the stress.

13. Some examples of stress-conditioned vowel alternation are given in 16.

(16) Stress-conditioned Vowel Alternations (Unstressed Vowel Raising) in Tocharian B

a. Stressed [ˈə] alternates with unstressed [ɨ] = /ə/
/tərkər/ → *tarkär* [ˈtərkɨr] 'cloud:NOM.SG'
vs. /tərkərwa/ → *tärkarwa* [tɨrˈkərwə] 'cloud:NOM.PL'

b. Stressed [ˈa] alternates with unstressed [ə] = /a/
/çərsa/ → *śarsa* [ˈçərsə] 'know:PST.3SG'
/çərsare/ → *śärsāre* [çɨrˈsare] 'know:PST.3PL'

14. Given that the vowel [a], which only surfaces under the stress, does not occur in final syllables (except when an underlying word-final /ə/ is deleted), and that an [a] or [ə] may be found in the first syllable of a disyllables, which then becomes [ə] or [ɨ], respectively, when a further syllable is added, final syllables must be treated as extrametrical.

(17) Stress Patterns of Tocharian B

a. /σ/: /ak/ → *āk* [ˈak] 'zeal:NOM.SG
b. /σ σ/: /ləkle/ → *lakle* [ˈləkle] 'pain:NOM.SG'
c. /σ σ σ/: /ləkle-ntə/ → *läklenta* [lɨkˈlentə] 'pain:NOM.PL'
d. /σ σ σ σ/: /enkək-e-mtər/ → *enkaskemtär* [eŋˈkəskemtɨr] 'take:PRES.1PL'
e. /σ σ σ σ σ/ /parak-əʂ̩ːenc͡ɕa/ → *parākäṣṣeñca* [pəˈrakɨʂ̩ːeɲc͡ɕə] 'prosper:CAUS.PART'
f. /σ σ σ σ σ σ/ /parak-əskemane/ → *parākäskemane* [pəˈrakɨskeməne] 'prosper:CAUS.PART';

- Compare *lakle* [ˈləkle] 'pain' versus *läklenta* [lɨkˈlentə] 'pains'. In general, barring cases where stress assignment is rendered opaque by processes of syncope, as well as some classes of morphologically motivated exceptional stress, primary stress falls on the second syllable of all words with three or more syllables.

- Compare the 5- and 6-syllable forms *parākäṣṣeñca* [pəˈrakɨʂːeɲc͡ɕə] 'prosper:CAUS.PART' and *parākäskemane* [pəˈrakɨskemǝne] 'prosper:CAUS.PART

15. Tocharian B thus appears to construct left-to-right iambs, with primary stress assigned to the leftmost foot.

- A difference in any of the parameters foot form, direction of parsing, or end rule would result in a different overall pattern, rather than consistent peninitial stress.
- Example (19) provides illustrative derivations for two-, three-, and six-syllable forms.

(18) Metrical Stress Parameters of Tocharian B

a. <u>Foot Form:</u> Iamb

b. <u>Direction of Parsing:</u> Left-to-Right

c. <u>End Rule:</u> Left

d. <u>Degenerate Feet:</u> Banned

e. <u>Syllable Extrametricality:</u> Yes

(19) Derivation of Primary Stress in Tocharian B

/ləkle/	/ləklentə/	/parakəskemane/	UR
lək.⟨le⟩	lək.len.⟨te⟩	pa.ra.kəs.ke.ma.⟨ne⟩	Final Syllable Extrametricality
(lək).⟨le⟩	(lək.len).⟨te⟩	(pa.ra).(kəs.ke).ma.⟨ne⟩	Foot Construction: Iamb, L-to-R
(ˈlək).⟨le⟩	(lək.ˈlen).⟨te⟩	(pa.ˈra).(kəs.ke).ma.⟨ne⟩	End Rule: Left
(ˈlək).⟨le⟩	(lɨk.ˈlen).⟨te⟩	(pə.ˈra).(kɨs.ke).mə.⟨ne⟩	Unstressed Vowel Raising
[(ˈlək).⟨le⟩]	[(lɨk.ˈlen).⟨te⟩]	[(pə.ˈra).(kɨs.ke).mə.⟨ne⟩]	SR

16. Footing in TB furthermore appears to be non-persistent.

- A process of schwa deletion (Pre-coronal Deletion, stated in (20); see Winter 1993, Kim 2011) may delete a schwa that would have been expected to bear primary stress, with the result that stress then instead falls on the initial syllable in affected forms, on a now monosyllabic foot.
- The derivations of inflected forms of the root /nek-/ 'disappear' illustrate apparent intraparadigmatic stress mobility between peninitial and initial stress due to Pre-coronal Deletion.

(20) Pre-coronal Deletion (cf. Koller 2021: 6)

$ə \rightarrow \varnothing$ / $V(C_i)C_j$ ___ $\begin{bmatrix}\text{+coronal}\\ \text{−sonorant}\end{bmatrix}$ V

where C_i, if present, is more sonorous than C_j

A vowel preceding a coronal obstruent immediately before another vowel is deleted if it is the second in a sequence of three vowels, and there is not a cluster of rising or flat sonority preceding the target vowel.

(21) Non-Persistent Footing in Cases of Schwa-Deletion[3]

/nek-ə-sta/	/nek-əsa-te/	UR
ne.kə.⟨sta⟩	ne.kə.sa.⟨te⟩	Final Syllable Extrametricality
(ne.kə).⟨sta⟩	(ne.kə).sa.⟨te⟩	Foot Construction: Iamb, L-to-R
(ne.ˈkə).⟨sta⟩	(ne.ˈkə).sa.⟨te⟩	End Rule: Left
(ne.ˈkə).⟨sta⟩	(ˈnek).sa.⟨te⟩	Pre-Coronal Deletion
(ne.ˈkə).⟨sta⟩	(ˈnek).sə.⟨te⟩	Unstressed Vowel Raising
[(ne.ˈkə).⟨sta⟩]	[(ˈnek).sə.⟨te⟩]	SR

- From the point of view of the comparative prosody of the (older) Indo-European languages, Tocharian B is somewhat remarkable in exhibiting iambic feet.

17. Here are the major constraints standardly employed in optimality-theoretic analyses of metrical stress systems. Definitions for these constraints are found at the end of the handout.

	Foot Structure	**Foot/Word Edge Alignment**	**Prosodic Parsing**	**Extrametricality**	**Syllable Rhythm**
	Iamb Trochee FtBin WSP Rh-Contour	All-Ft-L All-Ft-R Leftmost Rightmost Align-Wd-L Align-Wd-R	GrWd = PrWd Parse-σ Parse-Seg	NonFinality-σ NonFinality-Ϝ NonInitiality-Ϝ	*Clash
Total #	5	6	3	3	1

Table 1: OT Metrical Stress Constraints

18. Ranking considerations for an OT stress grammar of Tocharian B

   - An optimality-theoretic grammar of TB stress is comparatively simple, since the system is not quantity sensitive, meaning that constraints relating to quantity and foot form (WSP and Rh-Contour) may be ignored and placed in the lowest stratum.
   - Likewise, constraints that would prefer feet or stress closer to the right edge of the word (All-Ft-R, Align-Wd-R, Rightmost) are irrelevant and thus belong to the lowest stratum of the grammar.
   - Since NonInitiality-Ϝ is consistently violated by all winning candidates, it belongs to the lowest stratum as well.
   - Insofar as words of five or more syllables build multiple feet, e.g., [(pə.ˈra).(kɨs.ke).mə.⟨ne⟩] 'prosper:caus.part', All-Ft-L, too, can be placed in the lowest stratum, below Parse-σ.
   - This constraint and FtBin belong in a stratum together below NonFinality-σ and NonFinality-Ϝ, Parse-σ is consistently violated in order to avoid violation of these two constraints, and, as stress on the initial syllable in disyllabic forms like [(ˈlak).⟨le⟩] shows, degenerate monosyllabic feet are permitted to satisfy NonFinality.
   - Ranking above the NonFinality constraints, on the other hand, are the constraints satisfied by every winner, which always has a foot with primary stress at the left edge of the word (Align-Wd-L, Leftmost), is always iambic when disyllabic (Iamb), always has a primary stress, even in monosyllabic forms (GrWd = PrWd); *Clash and Parse-C are always untouched as well.

[3] Koller interprets the presence of primary stress on the initial syllable in cases like [(ˈnek).sə.⟨te⟩] where Pre-coronal Deletion has applied as evidence for both a need for iambic foot structure (as opposed to the foot-free analysis of Burroni 2017) and need for a derivational architecture, given that Pre-coronal Deletion opacifies the expected stress assignment (^{X}[(nek.ˈsə).⟨te⟩]). Although coda consonants generally do not contribute to syllable weight in TB, it is conceivable that the deletion of /ə/ creates a heavy syllable with moraic coda consonant in order to preserve the mora associated with /ə/ in in the input (Max-μ ≫ *Moraic Coda).

19. The total ranking is given in 22.

(22) Total Constraint Ranking for the Stress Grammar of Tocharian B:

GrWd = PrWd, Iamb, Align-Wd-L, Leftmost, *Clash, Parse-Seg ≫ NonFinality-σ, NonFinality-ƒ ≫ FtBin, Parse-σ ≫ All-Ft-L, All-Ft-R, Align-Wd-R, Rightmost WSP, Rh-Contour, NonInitiality-ƒ, Trochee

Constraint Definitions

(23) NonFinality-Foot (NonFin-ƒ): No foot is final the prosodic word. Assign one violation * if the right edge of a foot, ƒ, is aligned with the right edge of the prosodic word, ω.

(24) Parse-Consonant (Parse-C): All consonants should be parsed into syllables. Assign one violation * for each consonant that is not parsed into the next higher level prosodic category, i.e., the syllable (σ).

(25) Parse-Syllable (Parse-σ): All syllables should be parsed into feet. Assign one violation * for each syllable that is not parsed into the next higher level prosodic category, i.e., the foot (ƒ).

(26) FootBinary (FtBin): Feet should be binary at either the moraic or syllabic level. Assign one violation * for each foot in the output that does not dominate any branching constituent (i.e., all feet consisting of a single light syllable).

(27) Rh-Contour: A foot must end in a strong–weak contour at the moraic level. Assign one violation * for each trochaic foot of the shape (–́ ⏑) or iambic foot of the shape (⏑ ⏑́).

(28) Weight-to-Stress Principle (WSP): Heavy syllables should be stressed. Assign one violation * for each heavy syllable in the output that is unstressed (i.e., parsed into the weak part of a foot or not parsed into a foot at all).

(29) All-Feet-Left/Right (All-Ft-L/R):[4] Every foot should stand at the left/right edge of the prosodic word. Assign one violation * for each syllable that intervenes between the left/right edge of a foot and the left/right edge of the prosodic word.

(30) *Clash: No stressed syllables are adjacent. Assign one violation * for each pairwise continguous sequence of two stressed syllables.

(31) Leftmost/Rightmost:[5] The head foot should be leftmost/rightmost in the prosodic word. Assign one violation * for each syllable in the prosodic word that intervenes between the foot with primary stress and leftmost/rightmost syllable of the prosodic word.

(32) Align-(ω, Left/Right, Foot, Left/Right) (Align-Wd-L/R): Every prosodic word ω should begin/end with a foot. Assign one violation * the left/right edge of a prosodic word is not aligned with the left/right edge of a prosodic word.

(33) Trochee: Feet should be trochaic (= feet have initial prominence). Assign one violation * for each polysyllabic foot in the output which the strongest member is not leftmost in the foot.

(34) Iamb: Feet should be iambic (= feet have final prominence). Assign one violation * for each polysyllabic foot in the output in which the strongest member is not rightmost the foot.

[4] This constraint is a labeling convenience for a specific Align constraint, namely, Align-(Foot, Left/Right, ω, Left/Right); cf. Kager 1999: 163.

[5] Equivalent to Align-(Head-Foot, Left/Right, ω, Left/Right). These constraints are also labeled Edgemost in Prince and Smolensky 1993 [2004].

References

Allen, W. Sidney. 1972. *Sandhi*. The Hague: Mouton de Gruyter.

Arvaniti, Amalia, and Janet Fletcher. 2020. The Autosegmental-Metrical Theory of Intonational Phonology. In Carlos Gussenhoven and Aoju Chen (eds.), *The Oxford Handbook of Language Prosody*, 78–95. Oxford: Oxford University Press.

Bateman, Janet. 1990. Iau Segmental and Tone Phonology. *Nusa* 32.29–42.

Beguš, Gaspar. 2016. The Phonetics of the Independent Svarita in Vedic. In Stephanie W. Jamison, H. Craig Melchert and Brent Vine (eds.), *Proceedings of the 26th Annual UCLA Indo-European Conference*, 1–12. Bremen: Hempen.

Bennett, Ryan. 2012. Foot-conditioned Phonotactics and Prosodic Constituency. Ph.D. diss., University of California, Santa Cruz.

Blevins, Juliette. 1993. A Tonal Analysis of Lithuanian Nominal Accent. *Language* 69.237–73.

Bruce, Gösta. 1977. *Swedish Word Accent in Sentence Perspective*. *Travaux de l'Institut de Linguistique de Lund*, vol. 12. Lund: Gleerup.

Burroni, Francesco. 2017. Stressing the Evidence: An OT Account of Tocharian B Stress. Presentation. 29th Annual UCLA Indo-European Conference. November 3–4 2017.

Byrd, Andrew Miles. 2015. *The Indo-European Syllable*. Leiden: Brill.

Chafe, Wallace L. 1977. Accent and Related Phenomena in the Five Nations Iroquois Languages. In Larry M. Hyman (ed.), *Studies in Stress and Accent*, 169–81. Los Angeles: University of Southern California.

Dogil, Grzegorz. 1999. The Phonetic Manifestation of Word Stress in Lithuanian, Polish, German and Spanish. In Harry van der Hulst (ed.), *Word Prosodic Systems in the Languages of Europe*, 273–311. Berlin: Mouton de Gruyter.

Féry, Caroline. 2017. *Intonation and Prosodic Structure*. Cambridge: Cambridge University Press.

Golston, Chris. 1991. Both Lexicons. PhD Dissertation, University of California, Los Angeles.

Gordon, Matthew. 2002. A Factorial Typology of Quantity-Insensitive Stress. *Natural Language and Linguistic Theory* 20.491–552.

Gordon, Matthew, and Timo Roettger. 2017. Acoustic Correlates of Word Stress: A Cross-Linguistic Survey. *Linguistics Vanguard* 3(1).

Gunkel, Dieter. 2017. On the Accentual Status of Vocatives in Proto-Indo-European. URL `https://www.academia.edu/35102099/On_the_accentual_status_of_vocatives_in_proto_Indo_European`, Presentation. 29th Annual UCLA Indo-European Conference. November 3–4 2017.

Gussenhoven, Carlos. 2004. *The Phonology of Tone and Intonation*. Cambridge: Cambridge University Press.

Hayes, Bruce. 1995. *Metrical Stress Theory*. Chicago: University of Chicago Press.

Hellmuth, Sam. 2006. Intonational Pitch Accent Distribution in Egyptian Arabic. Ph.D. diss., School of Oriental and African Studies, University of London.

———. 2007. The Relationship between Prosodic Structure and Pitch Accent Distribution: Evidence from Egyptian Arabic. *The Linguistic Review* .

van Heuven, Vincent J. 2019. Acoustic Correlates and Perceptual Cues of Word and Sentence Stress. In Rob Goedemans, Jeffrey Heinz and Harry van der Hulst (eds.), *The Study of Word Stress and Accent. Theories, Methods and Data*, 15–59. Cambridge: Cambridge University Press.

van Heuven, Vincent J., and Alice Turk. 2020. Phonetic Correlates of Word and Sentence Stress. In Carlos Gussenhoven and Aoju Chen (eds.), *The Oxford Handbook of Language Prosody*, 150–165. Oxford: Oxford University Press.

Hualde, José Ignacio. 2000. On System-Driven Sound Change: Accent Shift in Markina Basque. *Lingua* 110.99–129.

———. 2012. Two Basque Accentual Systems and the Notion of Pitch-Accent Language. *Lingua* 122.1335–51.

Hyman, Larry M. 2006. Word Prosodic Typology. *Phonology* 23.225–57.

———. 2009. How (Not) to Do Phonological Typology: The Case of Pitch-Accent. *Language Sciences* 31.213–38.

———. 2014. Do All Languages Have Word Accent? In Harry van der Hulst (ed.), *Word Stress: Theoretical and Typological Issues*, 56–82. Cambridge: Cambridge University Press.

Kager, Rene. 1999. *Optimality Theory*. Cambridge: Cambridge University Press.

Kenstowicz, Michael. 1972. Lithuanian Phonology. *Studies in the Linguistic Sciences* 2.1–85.

Kim, Ronald. 2011. Eine indirekte Fortsetzung urindogermanischer Prosodie? Zweitsilbenakzent bei finiten Verben im Tocharischen. In Olav Hackstein and Ronald I. Kim (eds.), *Linguistic Developments along the Silk Road: Archaism and Innovation in Tocharian*, 9–32. Wien: Österreichischen Akademie der Wissenschaften.

Kisseberth, Charles W., and Mohammad Imam Abasheikh. 2011. Chimwiini Phonological Phrasing Revisited. *Lingua* 1987–2013.

Koller, Bernhard. 2021. Morphologically Conditioned Accent and Cyclic Derivation in Tocharian B. Presentation. Oxford Workshop on Indo-European Accentuation, 25 July 2021.

Krause, Wolfgang. 1952. *Westtocharische Grammatik. Band I. Das Verbum*. Heidelberg: Winter.

Krause, Wolfgang, and Werner Thomas. 1960. *Tocharisches Elementarbuch. Band I. Grammatik*. Heidelberg: Winter.

Krauss, Michael. 2005. Athabaskan Tone. In Sharon Hargus and Keren Rice (eds.), *Athabaskan Prosody*, 51–136. Amsterdam: John Benjamins.

Kuryłowicz, Jerzy. 1968. *Indogermanische Grammatik II. Akzent - Ablaut*. Heidelberg: Carl Winter.

Kushnir, Yuriy. 2018. Prosodic Patterns in Lithuanian Morphology. Ph.D. diss., Universität Leipzig.

Ladd, D. Robert. 1996. *Intonational Phonology*. Cambridge: Cambridge University Press.

Liberman, Mark. 1975. The Intonational System of English. Ph.D. diss., Massachusetts Institute of Technology.

Macdonell, Arthur Anthony. 1910. *Vedic Grammar*. Strassburg: Trübner.

Malzahn, Melanie. 2010. *The Tocharian Verbal System*. Leiden: Brill.

McCawley, James D. 1978. What is a Tone Language? In Victoria Fromkin (ed.), *Tone: Linguistic Survey*, 113–31. New York: Academic Press.

Melinger, Alissa. 2002. Foot Structure and Accent in Seneca. *International Journal of American Linguistics* 68.287–315.

Mester, R. Armin. 1994. The Quantitative Trochee in Latin. *Natural Language and Linguistic Theory* 12.1–61.

Pierrehumbert, Janet. 1980. The Phonology and Phonetics of English Intonation. Ph.D. diss., Massachusetts Institute of Technology.

Pinault, Georges-Jean. 2008. *Chrestomathie tokharienne. Textes et grammaire*. Leuven - Paris: Peeters.

Prince, Alan, and Paul Smolensky. 1993 [2004]. *Optimality Theory: Constraint Interaction in Generative Grammar*. Wiley–Blackwell. URL `http://roa.rutgers.edu/files/537-0802/537-0802-PRINCE-0-0.PDF`.

Riad, Tomas. 2014. *The Phonology of Swedish*. Oxford: Oxford University Press.

Winter, Werner. 1993. The Development of Underlying Accented "Schwa" before Dental in Tocharian. *Tocharian and Indo-European Studies* 6.197–205.

Witzel, Michael. 2020. Zu den vedischen Akzent-Systemen. In Norbert Oettinger, Stefan Schaffner and Thomas Steer (eds.), *„Denken Sie einfach!" Gedenkschrift für Karl Hoffmann*, 255–84. Dettelbach: Röll.

Yip, Moira. 2002. *Tone*. Cambridge: Cambridge University Press.

Young, Steven R. 1991. *The Prosodic Structure of Lithuanian*. Lanham, MD: University Press of America.